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Scam detection between individuals with and without prior victimization

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ABSTRACT

Despite extensive research on phishing susceptibility and training effectiveness, a critical gap remains in understanding the differences in scam detection ability between individuals with and without prior victimization experience. Towards this, we assessed the scam detection skills of 117 participants using phone-based and email-based scenarios. Although results showed no links between victimization status and demographics, it demonstrated that previously victimized individuals perform significantly worse at identifying phone-based malicious scenarios compared to non-victimized participants. Notably, individuals with prior victimization experience did not show higher confidence in their scam detection abilities or greater perceived helpfulness of past experiences compared to those without such experiences. Qualitative analysis revealed that errors stemmed primarily from excessive trust, disproportionate focus on immediate loss, and over-reliance on similar past experiences. These findings suggest that prior scam victimization does not necessarily lead to higher resilience against scams and underscore the need for targeted interventions in scam-prevention education to calibrate individuals' trust bias, reduce excessive focus on immediate loss, and address over-reliance on superficially similar past experiences. Implications of our findings are discussed in the paper.

1. Introduction

The proliferation of digital communication has created unprecedented opportunities for cybercriminals to exploit human vulnerabilities through various forms of scams. In this paper, we use the term “scam” to refer to any deceptive and fraudulent activities (e.g., phishing attack, fake online deals) that attempt to trick individuals for financial gains and/or to get access to their personal information/data. Such scams may occur on the Internet, over the phone/video calls, on social media apps (e.g., Discord, TikTok), or via messaging applications (e.g., Facebook Messenger, Skype). Recent data reveals the staggering scale of this threat, with over 6.47 million fraud reports filed in 2024 alone, resulting in \$12.5 billion in losses (Network, 2025). Over the years, phishing attacks have shown remarkable resilience and adaptability, with 932,923 attacks recorded in the third quarter of 2024 alone, a 6.3% increase from the previous quarter (APWG, 2024). This persistent threat landscape has prompted extensive research into both the technical and human factors. While technological defenses leveraging sophisticated machine learning approaches are reported to achieve detection accuracies exceeding 90% (He et al., 2023; Mughaid et al., 2022), human vulnerability remains the weakest link in cybersecurity. Recent research reported that 74% of security breaches involve human error, where scams are specifically designed to exploit cognitive biases, emotional triggers, and behavioral patterns (Arévalo et al., 2023;

Desolda et al., 2021). Thus, understanding how individuals detect and respond to various scams is important and requires examining the nuanced, complex interplay among expertise, cognitive processing, and past experiences.

A large volume of prior efforts has examined the human factors related to scams in general. In this vein, Wash (2020) have shown that successful phishing detection relies heavily on contextual expertise and the ability to detect discrepancies, rather than purely on technical knowledge. Even for non-experts, it is shown that they bring unique capabilities to phishing detection, including contextual knowledge about email expectations and sender history that automated systems cannot always replicate (Wash et al., 2021). Beyond email, sophisticated voice-based attacks present unique challenges due to real-time adaptation and the absence of traditional metadata cues (Derakhshan et al., 2021). Prior efforts have also examined various forms of scam-detection training and tested their efficacy (Weaver et al., 2021; Jensen et al., 2017; Zheng and Becker, 2023; Hull et al., 2023; Singh et al., 2023).

Despite extensive research on factors related to scam susceptibility and training effectiveness, a critical gap remains in understanding how individuals with prior victimization experiences respond differently to potential threats than those without such experiences. While some research has examined repeat victimization patterns (DeLiema et al., 2024; Whitty, 2019), it remains unclear how these two groups differ and whether victimization experiences could fundamentally alter

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individuals' cognitive processing, emotional responses, behavioral patterns, and risk mitigation abilities when encountering threats. In a related vein, [DeLiema et al. \(2025b\)](#) recently showed that there is no association between prior scam victimization and lower trust in government or in online transactions, challenging assumptions about how victimization shapes subsequent attitudes. This finding raises important questions about whether victimization experiences translate into productive skepticism that enhances detection ability or whether victims maintain vulnerability patterns that persist despite their experiences.

To address this critical gap, the present study examines differences between individuals with prior experience of scam victimization and those without such experience. Unlike previous research that has either excluded victims or failed to account for victimization history, we explicitly compare these groups and aim to answer the following research questions:

- **RQ1.** Do participants with scam victimization experience exhibit greater accuracy in detecting **scams** compared to individuals with no history of victimization?
- **RQ2.** Do participants with scam victimization experience exhibit greater accuracy in detecting **legitimate** communication compared to individuals with no history of victimization?
- **RQ3.** Do participants with scam victimization experience differ while labeling **ambiguous** scenarios as scams (or legitimate) compared to individuals with no history of victimization?
- **RQ4.** Do participants with scam victimization experience exhibit increased confidence in detecting scams compared to individuals with no history of victimization?
- **RQ5.** Do participants with scam victimization experience perceive their past experiences as beneficial while identifying scams compared to individuals with no history of victimization?

To answer these questions, we designed a user study consisting of two phases. In Phase 1, we collected responses to screen out participants with any cybersecurity training experience, to avoid sample bias. Phase 2 (main study) was divided into four parts. The first part focused on collecting information about participants' prior experiences of scam victimization. The second part included 12 tests, both phone- and email-based, comprising 6 scams, 4 legitimate, and 2 ambiguous scenarios. The third part aimed to explore participants' reasoning for each test case, their confidence in those decisions, and their perceptions of how helpful their past scam experience was in making them. Finally, the fourth part collected demographic information, including age, gender, income, and education level.

Our analyses revealed no significant associations between victimization status and age, gender, income, or education. However, the results demonstrated that previously victimized individuals performed significantly worse at identifying phone-based scam scenarios than those with no prior victimization experience. Notably, individuals with prior scam victimization did not report greater confidence in scam detection or higher perceived helpfulness of past experiences than those without such experiences.

2. Background

2.1. Factors relevant to scam detection ability

A large body of work has examined various factors that may influence individuals' ability to detect scams. In this vein, several demographic and psychological predictors have been identified as relevant factors in phishing vulnerability, including risk-taking behaviors, decision-making styles, and personality traits ([Abroshan et al., 2021b](#)). Personality assessments reveal that individuals with high influence traits (i.e., outgoing, people-oriented, sociable, talkative) show lower susceptibility, while those with dominant (i.e., outgoing, task-oriented, bold, fearless) or steadiness traits (i.e., reserved, people-oriented, generous, patient) exhibit higher vulnerability ([Sumner et al., 2022](#)), and

elevated emotional arousal compromises consumer vigilance across all age groups, regardless of whether the emotional state is positive or negative ([Kircanski et al., 2018](#)).

Prior efforts have also examined the decision-making process for scam detection. Specifically, prior research has identified a three-stage detection process: the initial sensemaking stage, where communications are interpreted at face value; the triggering of suspicion based on accumulated discrepancies; and the decision-making stage ([Wash, 2020](#)). Further, a rational decision-making style is associated with reduced vulnerability, whereas an intuitive and spontaneous style is associated with increased vulnerability. Successful detection is found to rely heavily on contextual expertise and an understanding of the communication style/pattern of legitimate organizations, rather than purely on technical knowledge ([Abroshan et al., 2021b](#)). In addition, the SCAM model (Suspicion, Cognition, and Automaticity) demonstrates that heuristic processing and automatic email habits significantly increase phishing susceptibility while systematic processing involving effortful scrutiny reduces it ([Vishwanath et al. \(2018\)](#)). This automaticity effect creates a paradox where frequent Internet users become more vulnerable through habituation despite greater exposure to security warnings ([Ribeiro et al., 2024](#)).

In addition to cognitive and habitual processes, triggered emotions can bypass rational decision-making and increase susceptibility to scams. Specifically, experimental research has shown that both positive emotions (e.g., excitement) and negative emotions (e.g., anger, fear) can significantly increase susceptibility to fraudulent advertisements compared to low-arousal states ([Kircanski et al., 2018](#)). Furthermore, context-specific emotional states have been shown to exacerbate this condition. For instance, during the COVID-19 pandemic, fear related to COVID-19 increased susceptibility to pandemic-themed scams ([Abroshan et al., 2021a](#)). However, the relationship between emotion and vulnerability is not one-sided. Specifically, while negative moods may trigger more systematic processing and reduce vulnerability, the mood-maintenance view suggests that individuals in a positive mood may avoid information that might disrupt their current state, thereby increasing vulnerability ([Norris and Brookes, 2021](#)). This theoretical framework helps explain why emotionally-charged scam attempts, particularly those exploiting authority bias or urgency tactics, remain highly effective ([Quinkert et al., 2021](#)). In addition to emotion, self-efficacy in threat detection can be influenced by observational, heuristic, and structural knowledge ([Baral and Arachchilage, 2019](#)). Besides the aforementioned factors, Fear of Missing Out (FoMO) is identified as a strong predictor, with high-FoMO individuals exhibiting poorer security knowledge and riskier behaviors ([Hadlington et al., 2020](#)).

Finally, prior efforts have extensively examined the effectiveness of anti-scam training, reporting mixed results regarding long-term retention and behavioral change. While various training approaches, including rule-based instruction ([Weaver et al., 2021](#)), mindfulness techniques ([Jensen et al., 2017](#)), adversarial mindset training ([Zheng and Becker, 2023](#)), and narrative-based method ([Hull et al., 2023](#)), have shown promise in improving short-term detection ability, evidence suggests these gains often decay within weeks ([Marshall et al., 2024](#); [Lastdrager et al., 2017](#)). However, while detailed feedback after training improves sensitivity, excessive exposure to scam examples can paradoxically increase false alarms and overconfidence ([Singh et al., 2023](#)). These prior results suggest that further research is needed to understand the nuanced interactions among different factors that contribute to long-term scam susceptibility.

2.2. Impact of prior victimization

Research on repeat victimization demonstrates that prior victims face an elevated risk of subsequent targeting and exploitation. In this vein, [DeLiema et al. \(2024\)](#) analyzed perpetrators' customer relationship management data seized by the USPIS (U.S. Postal Inspection

Service) from mail scams involving 1.33 million U.S. victims, finding that victims aged 70–89 had a 9% higher risk of repeat victimization compared to those in their 50s. Notably, the time between victimization incidents decreased as prior victimization increased, with shorter intervals between the 10th and 11th incidents than between the 1st and 2nd. This suggests that initial victimization is associated with future vulnerability that compounds over time rather than generating protective learning effects. It could be due to a number of reasons, such as shifts in underlying mental models due to repeated victimization, and/or certain flaws in the decision-making process of the victims that remain unchanged over time. A later work by DeLiema et al. (2025b) supports this to some extent, finding no association between prior scam victimization and lower trust in the government, the Social Security Administration, or online transactions. Interestingly, Whitty (2019) noted that repeat victims show minimal socio-demographic differences compared to one-off victims, further suggesting that decision-making processes and cognitive factors are more likely to explain vulnerability.

Notably, prior efforts did not investigate whether people with prior victimization differ in their ability to detect scams from those with no prior victimization, and defined victimization solely in terms of financial loss, excluding data loss. DeLiema et al. (2023) focused on intra-category susceptibility (e.g., why a person targeted by a consumer purchase scam responds to that specific scam) rather than examining whether and why prior scam victims are more vulnerable than people who have not been victimized. In addition, examining whether recent scam encounters shape users' perceived susceptibility, as measured by asking participants how likely they are to become victims of such attacks (Chen et al., 2020), may not reflect users' actual vulnerability.

We argue that, rather than focusing solely on self-reported measures of scam susceptibility, it is important to investigate whether victims of scams differ from non-victims in their decision-making and cognitive factors, as reflected in their objective detection ability. Specifically, victims who recognize and understand how they were deceived might develop enhanced detection ability, while those who remain unaware of deception tactics and/or blame themselves might remain vulnerable. The absence of systematic research comparing detection ability between victims and non-victims represents a significant gap in understanding these dynamics, and is addressed by our work.

3. Methodology

The study presented in this paper is designed as a two-phase survey. The Phase 1 survey asks participants about their prior cybersecurity training. Based on responses from Phase 1, we excluded participants who reported receiving cybersecurity training from the Phase 2 survey. We did this to avoid potential confounding effects, as trainings can vary substantially in format, depth, content, and recency. Treating it as a binary variable would mask this heterogeneity.

The Phase 2 survey aims to examine the invited participants' past experience with scams and their ability to detect scams using the designed tests. After collecting data in Phase 2, we grouped final responses into two groups based on participants' prior experience as scam victims. For participants who had been victimized by at least one scam and experienced financial, material, and/or personal information/data losses, we identified them as the victim group (referred to as the "victimized" group from here on). The remaining participants who had not been victimized by any scam in the past were identified as the non-victim group (referred to as the "non-victimized" group from here on). Regarding grouping, the victimized group includes only those who suffered measurable harm (e.g., financial, material, and/or data losses). This ensures that the victimized group includes participants who experienced real-world losses, avoiding cases with no actual loss (e.g., attempted frauds that failed). On the contrary, the non-victimized group acts as a control to isolate the effects of true victimization as best as possible.

3.1. Informed consent

The study was conducted via an online survey hosted on Qualtrics. Participants were required to review an electronic consent form and provide informed consent electronically before beginning the survey.

3.2. Participant requirement

All participants were recruited through the Prolific platform. We implemented several precautionary measures to ensure high-quality data and thoughtful responses. *First*, we recruited participants who were at least 18 years old, currently residing in the United States, and whose first language was English. In addition, eligibility was limited to individuals with an approval rate of at least 95% and at least 50 prior submissions. *Second*, to ensure that no one bypassed our eligibility criteria or used bots to complete the survey, participants were required to enter their unique Prolific ID before accessing the survey questions. These IDs were stored in the Prolific platform, allowing us to verify that each participant could complete the survey only once. *Third*, multiple attention-check questions were included throughout the survey. Participants were excluded from the analysis if they answered any attention check question incorrectly (e.g., selected any answers other than "Moderately Severe" for Q3 or selected any answers other than "True" for Q4 in the Phase 2 survey). The inclusion criteria (i.e., age ≥ 18 , U.S. resident, native English speaker, approval rate $\geq 95\%$, ≥ 50 prior submissions, one-time participation) were consistently applied across both phases of the study, whereas attention checks were used in Phase 2. Additionally, only a subset of participants from Phase 1 was invited to participate in Phase 2 based on their lack of cybersecurity training experience. To minimize sample bias, participants who reported having no prior cybersecurity training were randomly selected for Phase 2.

3.3. Survey flow

After providing consent for the Phase 1 survey, participants answered four prescreening questions to determine their eligibility. Eligible participants were then randomly selected and invited to participate in the Phase 2 survey. Upon consenting to the Phase 2 survey, participants reported their past experiences with scam victimization, including the impacts and their related knowledge. Next, participants completed a series of scam detection tests, which included six phone-based and six email-based scenarios. The tests were presented in audio and image formats. For each test, we included manipulation check questions (e.g., "Which of the following is true about the phone call? You must answer this correctly before you can move to the next question.") to ensure that participants carefully listened to and/or read the test materials before they were allowed to answer whether the scenario was fraudulent or legitimate. Finally, the Phase 2 survey concluded with participants completing a demographic questionnaire.

The Phase 1 survey took approximately 2.3 min to complete ($M = 2.318$, $Mdn = 1.733$, $SD = 1.97$), while the Phase 2 survey required about 52.5 min ($M = 52.513$, $Mdn = 47.883$, $SD = 21.955$). We paid \$0.50 USD for completing Phase 1 and \$9 USD for completing Phase 2 of the survey. Both surveys were hosted on the University of Connecticut's Qualtrics server and were approved by the Institutional Review Board (IRB).

3.4. Design of the test cases

To evaluate participants' ability to detect scams, we created three categories of test scenarios, namely, fraudulent, legitimate, and ambiguous. All scenarios were collected from either author's real experiences, websites of legitimate entities (ScammerRevolts, 2024; ModderPaulScamBaits, 2024), or other studies (TalkBank, 2024; Wu et al.,

2020). These scenarios were presented in two formats: phone-based and email-based as follows:

Fraudulent (Scam) scenarios involved deceptive attempts in which correct identification by a participant was classified as a True Positive, and failure to identify them was categorized as a False Negative.

Legitimate scenarios depicted legitimate communications; accurate identification in these cases was marked as a True Negative, while misidentification was labeled as a False Positive.

Ambiguous scenarios could be interpreted as either legitimate or a scam, depending on the interpretation of the communication and the missing (or included) cues. These scenarios were intentionally ambiguous, lacking clear indicators of malicious intent, and were open to interpretation by participants. The goal of including these was to elicit participants' responses that can provide important insights into their decision-making process.

The fraudulent and legitimate scenarios were designed to evaluate participants' fundamental scam-detection ability, while the ambiguous scenarios aimed to explore how prior experience with scams influenced participants' levels of suspicion.

In phone-based communication scenarios, to eliminate biases related to accents and racial perceptions, we used Adobe Premiere Pro (Adobe Inc., 2024) and ElevenLabs (ElevenLabs Inc., 2024) to standardize audio test accents. In all cases, the accent of the person who is likely to be the target of suspicion (irrespective of who initiated the call) was adjusted. All accents were standardized to a uniform British female accent to ensure consistency across the tests. We informed participants about this as follows: "One of the voices is replaced with an AI-generated voice to avoid biasing your response."

3.4.1. Design of the scam/fraudulent test scenarios

We designed a total of six scam scenarios, each based on different types of scams and distinct techniques (i.e., Test 2, Test 4, Test 6, Test 7, Test 10, and Test 12 throughout this discussion; all tests are presented in the Supplementary Material). Tests 2, 6, and 10 depict scam scenarios related to consumer purchase, Test 4 focuses on a threat-based scam, and Tests 7 and 12 represent phishing scams. The following sections outline the specific characteristics of each scenario, emphasizing the key indicators of fraudulent activity.

- **Test 2 (Phone-based): Vishing Impersonating PayPal Support (ScammerRevolt, 2024).** This scenario is designed as a vishing attack where an attacker impersonates a PayPal representative. The scenario begins with a user reporting a suspected fraudulent invoice received via email. The attacker was immediately informed about the compromised account security and the creation of a fraudulent account, employing tactics designed to induce panic. Critical indicators include rapid assertion of compromise without investigation, vague language (e.g., "suspicious activities going on with your information"), and pressure tactics.
- **Test 4 (Phone-based): Vishing Impersonating Wells Fargo Support (ModderPaulScamBaits, 2024).** This scenario involves a vishing attack mimicking Wells Fargo customer support, triggered by a text message reporting an unauthorized Bitcoin purchase. The user expresses concern regarding their inability to legally purchase cryptocurrency. The attacker claims to have identified multiple fraudulent accounts linked to the user and offers a refund that requires connecting to a "Wells Fargo secure server". A significant red flag is the request to connect to a separate, unverified "secure server", as legitimate banking representatives would never employ this tactic. Furthermore, the attacker fails to verify the caller's identity before discussing account details.
- **Test 6 (Phone-based): Combination of SMS messaging and Vishing Impersonating Listerhill Bank Support.** This scenario involves a combination of SMS communication followed by a phone call. The user first receives a text message about a potential Walgreens

purchase, prompting them to reply. Once the user responded with a "No", they received another text stating that their card has been restricted, followed immediately by a phone call. During the call, an individual claiming to be a representative from the Listerhill Bank alleges possible card theft. The caller (scammer) offers to cancel or replace the card but requires the user to provide Personally Identifiable Information (PII). Key indicators of fraudulent activity in this scenario include the rapid sequence of an automated text followed by a phone call that bypasses standard verification procedures, and the request for the user's full Social Security number. Also, by sending the SMS message first and informing the recipient of the follow-up call, the scammer makes the call appear as "expected," possibly causing the end user to let his/her guard down.

- **Test 7 (Email-Based): Zelle Fraud Attempt.** This scenario involves a follow-up email falsely claiming to be a payment confirmation from Zelle. After the recipient sold an item to the scammer on Facebook Marketplace, the email claims that \$220 has been received but is being held for verification because the recipient does not have a business account. The user is instructed to request an additional \$380 from the scammer to unlock the full amount. Indicators of fraud in this scenario include the use of a Gmail address ("zellebusinessserver@gmail.com") as the sender's email, which is not an official Zelle domain address. The requirement to send additional money to receive funds is a common feature of fraudulent schemes. Additionally, awkward phrasing, grammatical errors, and an overall unprofessional email design are key signs of deception.
- **Test 10 (Email-Based): PayPal Fraud Attempt.** This scenario involves a fraudulent payment confirmation email claiming a \$985.91 charge from PayPal. The email includes invoice details and a customer support phone number. The key red flags in this case include the sender's email address ("spdtmberget7@gmail.com"), which originates from a personal Gmail account rather than an official PayPal domain. Another unusual detail is the inclusion of recipient information within an invoice sent to the payer. Additionally, the email provides a non-official PayPal support number, creates a false sense of urgency by stating a 24-hour deadline, and contains poor grammar and an unprofessional design.
- **Test 12 (Email-Based): Cisco VPN Update Scam.** This scenario involves an email falsely claiming to be from Cisco, urging the recipient to take immediate action regarding a security update for the AnyConnect VPN client. The email warns of potential vulnerabilities and mentions the possibility of credential requests during the update process. Indicators of fraud in this scenario include the use of a non-official domain that includes words such as "cisco" to make it appear legitimate ("cisonetworks-solutions"), the use of imperative language designed to create a sense of urgency or panic, the absence of a personalized greeting, and the expectation that users must re-enter their login credentials during the update process, which is atypical for legitimate updates.

3.4.2. Legitimate test scenarios

The legitimate tests consist of Test 3 (Phone-Based) and Tests 8, 9, and 11 (Email-Based).

- **Test 3 (Phone-Based)** involves a casual phone conversation between friends discussing recent life updates (Wu et al., 2020).
- **Test 8 (Email-Based)** features an email announcing a new multi-year partnership between the Detroit Pistons and StockX, with StockX becoming the official jersey sponsor for the team starting in the upcoming season.
- **Test 9 (Email-Based)** includes an email informing recipients that single-game tickets for the 2024–25 Atlanta Hawks home games are now available for purchase, encouraging them to secure tickets for the upcoming season.

- *Test 11 (Email-Based)* presents an email notification from a school's IT Services, requesting verification of the password-recovery information associated with the recipient's ID. It confirms the accuracy of the phone number and email address on record.

3.4.3. Ambiguous tests

The ambiguous tests include Test 1 and Test 5 (TalkBank, 2024).

- *Test 1 (Phone-Based)* involves a short phone call between an individual identifying as a representative of Harper Valley National Bank and a customer. During the call, the customer requests and completes a transfer of \$135 from their savings account to their checking account. This scenario can be interpreted as fraudulent due to the lack of verification measures (e.g., confirming the last four digits of the Social Security Number, verifying the address, or using a security code sent via text). However, it can also be interpreted as legitimate because the individual transfers a small amount of \$135 between their own accounts, which could indicate a simplified process. Scammers often attempt larger transfers to maximize their gains, but this alone is not a definitive indicator of fraud.
- *Test 5 (Phone-Based)* features a phone call between an individual identifying as an Amazon representative and a customer. The customer requests a password reset, provides her phone number, and the representative confirms that a password reset link has been sent to that number. This scenario can be interpreted as fraudulent because a more secure and convenient method to reset passwords is available directly on the Amazon website. Additionally, the lack of robust verification introduces a potential risk of phone takeover, as anyone with access to the linked phone number could reset the password. However, this scenario can also be interpreted as legitimate in certain contexts, such as when elderly users, who may be less familiar with online password management, find it easier to request assistance over the phone. Furthermore, without considering the phone takeover risk, sending a password reset link to the linked phone number is common practice, as phone numbers can be used to verify the caller's identity. Notably, assuming a phone is always in the account holder's possession introduces a novel vulnerability to SIM-swap attacks that can take over people's online accounts.

3.5. Variables in the study

3.5.1. Independent variable

In this study, the independent variable, namely, past victimization status, was measured using two questions. "Have you ever experienced any financial or material losses (e.g., smartphones, laptops, or anything else that can be sold for money) due to scams?" and "Have you ever experienced loss of personal information/data (e.g., name, date of birth, SSN, credit card number, etc.) due to scams?". Participants are assigned to the victimized group if they answered "Yes" to either of these two questions. In contrast, those who answered "No" or "I don't know" to both questions are assigned to the non-victimized group. We also measured participants' demographic variables (e.g., Age, Income, level of education) at the end of the survey. Due to space limitations, we list only the independent variables analyzed in this paper.

3.5.2. Dependent variable

In this study, we calculated the accuracy score for Tests 1–12 individually as follows. For each test, participants were asked to listen to and/or read the corresponding materials and answer the question: "How likely is this telephone call/email to be a scam?" Responses were rated on a scale from 1 to 5, where 1 = "Not a scam at all", 2 = "Most likely not a scam", 3 = "Might be a scam", 4 = "Very likely to be a scam", and 5 = "Definitely a scam". As we have both scam and

legitimate test scenarios, the **accuracy scores** were calculated differently depending on the test's legitimacy. For *scam tests*, the accuracy score matched the order of the options (1 to 5). For *legitimate tests*, the accuracy score followed the reverse order of the options (5 to 1) (e.g., If a participant selects "Not a scam at all" for a scam scenario, the participant receives an accuracy score of 1 for this scenario; if a participant selects "Not a scam at all" for a legitimate scenario, the participant receives an accuracy score of 5 for the corresponding scenario).

In addition to individual scores for each test case, average scores were calculated for different types of test scenarios, including Ambiguous Scenario Scores, Phone-Based Fraudulent Scores, Phone-Based Legitimate Scores, Email-Based Fraudulent Scores, and Email-Based Legitimate Scores, by averaging the scores of the corresponding tests.

3.5.3. Data preprocessing

Regarding participants' responses to each test, we anticipated that some individuals might misunderstand the test materials. For example, in Test 4, the scammer uses urgency and phishing tactics (e.g., referencing a "secure server") to deceive the caller into providing sensitive information or money. Some participants perceived the phone call itself as legitimate but believed the scammer's description of an urgent situation was fraudulent. These participants mistakenly thought the scammer was helping the caller and marked this test as a scam for the wrong reason. Ideally, they should have recognized that the call itself was a scam and initiated by a scammer. As such, based on their reasoning, these participants would likely fall victim to such scams in real-life scenarios, especially if the scammer does not ask to connect to a secure server and uses some other forms of scams. To identify such cases and validate participants' understanding of each test, we reviewed their responses to the open-ended question "Briefly describe your reasoning behind your answer", which followed the question "How likely is this telephone call/email to be a scam?" We checked for consistency between their selected rating and their explanation. For example, if a participant answered "Definitely a scam" but provided reasoning that contradicted this response, we marked the code "Wrong Understanding/Unmatched Answer". Otherwise, we left the code blank. This process ensured that responses were consistent and allowed us to remove inconsistent ones. Additionally, the email used in Test 12 was a simulated phishing message from KnowBe4, a security awareness company that helps organizations test their employees with simulated fraudulent tests. While the email content referenced Cisco, the sender domain (psm.knowbe4.com) revealed its KnowBe4 origin. One participant identified the email as "Most likely not a scam" solely on the basis of recognizing the KnowBe4 domain, without engaging with the email's actual content or fake sender domain. Because this judgment did not reflect genuine reasoning for scam detection, we excluded this participant's Test 12 data from the analysis.

3.5.4. Data analyses

First, to explore whether demographic factors differ significantly between the victimized and non-victimized groups, we compared age, gender, income, and education. Mann–Whitney U tests were applied to age, income, and education due to non-normality, and gender was assessed using a Chi-square test for male/female categories, and Fisher's Exact Test for male/female/non-binary participants.

Second, to thoroughly analyze the experimental results, we applied a two-step analytical approach. We categorized responses according to performance in each test scenario. For fraudulent and legitimate scenarios, each response was labeled as either "Correct" or "Incorrect" based on its accuracy relative to the ground truth. For ambiguous scenarios, responses were categorized as "Non-Scam" or "Scam" to examine potential differences in distribution between the victimized and non-victimized groups. We then conducted Chi-Square tests to assess the statistical significance between victimization status (i.e., Victimized vs. Non-Victimized) and response accuracy (i.e., Correct vs. Incorrect).

Additionally, we analyzed the relationship between victimization status and response categories (i.e., “Non-Scam” vs. “Scam”) in ambiguous scenarios to determine whether the level of suspicion of fraud was associated with prior victimization.

Given the characteristics of the 12 tests, which assess related facets of the same underlying ability (scam detection ability), we cannot assume that these measures are independent. Applying familywise corrections such as Bonferroni would therefore be overly conservative and would substantially increase the risk of Type II errors (i.e., failing to detect genuine differences between victimized and non-victimized participants when such differences actually exist because the corrected p -value threshold is too stringent) (Shi et al., 2012; Ozenne et al., 2023; Armstrong, 2014; Vickerstaff et al., 2019). For these reasons, we report uncorrected p -values alongside effect sizes for significant results to provide a transparent and interpretable account of the findings.

Third, to complement the categorical analysis and minimize potential biases from pre-defined groups, we conducted non-parametric Mann–Whitney U tests on the raw accuracy scores for Tests 1 through 12. These tests were performed for each individual test, for combinations based on scenario types, and for false positive responses (i.e., incorrectly classifying legitimate scenarios as scams) and false negative responses (i.e., incorrectly classifying scam scenarios as legitimate). This approach provided a more detailed understanding of cognitive vulnerabilities under different conditions.

Finally, to better understand why participants in the victimized and non-victimized groups performed differently across tests, we conducted a qualitative analysis of their responses to the open-ended question, “Briefly describe your reasoning behind your answer”, for each test.

3.6. Qualitative analysis

3.6.1. Codebook development

Our qualitative data came from an open-ended question in the Phase 2 Survey Questionnaire (Question 3: “Briefly describe your reasoning behind your answer.”), which followed each test scenario. We received a total of 1404 open-ended responses to work with.

We analyzed participants’ open-ended responses using inductive qualitative content analysis, in which codes were developed directly from participants’ written reasoning (Hsieh and Shannon, 2005; Elo and Kyngäs, 2008). Because our research question concerned how victimized and non-victimized participants reasoned about correct vs. incorrect responses (and scam vs. non-scam classification in ambiguous scenarios), we constructed 24 codebooks in total.

3.6.2. Coding procedure

The first author reviewed all participant responses within each codebook and inductively generated codes that captured the rationales expressed (e.g., No Personal Data/Money Request, Suspicious Invoice). This iterative review produced 150 codes across the 24 codebooks (refer to Table 7 in Appendix A for the complete code table).

For coding, the first author initially reviewed all participant comments (refer to responses to Question 3) in each codebook to identify applicable and independent codes by examining participants’ reasoning for their decisions. Following this, the first author and a second coder independently evaluated each comment in every codebook. To ensure the validity of every response, if a comment was inconsistent with the participant’s answer (e.g., a participant selected “Not a scam at all”, but his/her comment suggested otherwise) or indicated a misunderstanding of the test material, or expressed uncertainty about their own judgment (e.g., “...not sure it was a scam.”, “I don’t know”) the code “Wrong Understanding/Unmatched Answer/Not Sure” was marked, and excluded from the analysis for the corresponding test scenario. Otherwise, this field was left blank, and other relevant codes (e.g., No Personal Data/Money Request, Suspicious Invoice, etc.) were noted as appropriate. After completing all 24 codebooks, the two coders met to resolve any discrepancies. Once conflicts were addressed, both coders independently updated their codebooks.

3.6.3. Inter-rater reliability

Upon the completion of updates, Inter-rater reliability (IRR) for each codebook was calculated using Cohen’s Kappa (McHugh, 2012). Due to unbalanced marginals with proportional randomness (we expected a majority of participants to understand each test material correctly and answer consistently) (Delgado and Tibau, 2019), percentage agreement ($\frac{\text{Total Match}}{\text{Total Counts}}$) was calculated to make sure agreement is valid. Among the 24 codebooks (150 codes) eligible for Cohen’s Kappa calculation, 105 codes (70.00%) demonstrated almost perfect agreement ($0.81 \leq \kappa \leq 1.00$), 27 codes (18.00%) showed substantial agreement ($0.61 \leq \kappa \leq 0.80$), 5 codes (3.33%) exhibited moderate agreement ($0.41 \leq \kappa \leq 0.60$), 6 codes (4.00%) indicated fair agreement ($0.21 \leq \kappa \leq 0.40$), and 1 code (0.67%) indicated slight agreement ($0 < \kappa \leq 0.20$). A total of 6 codes (4.00%) had a Kappa score of zero due to a high level of raw agreement (Byrt et al., 1993); for these 6 codes, the percentage agreements are in range between 97.22% and 99.02%, reflecting an almost perfect level of agreement. For all 150 codes, the percentage agreements of $\geq 70.69\%$ indicate substantial agreement between coders overall.

3.7. Survey data analysis

We collected a total of 500 responses for the Phase 1 survey. Of these, 5 responses were excluded due to inconsistencies. Among 204 participants with prior cybersecurity training, 45.6% ($n = 93$) reported being victims of at least one scam, while 54.4% ($n = 111$) indicated no such victimization experience. In contrast, within the group without cybersecurity training ($n = 291$), 47.8% ($n = 139$) reported being victims of at least one scam, while 52.2% ($n = 152$) had not experienced any scams. After Phase 1, we randomly selected and invited 270 participants from the group without cybersecurity training to participate in the Phase 2 survey. Of these, 118 participants completed the survey, and 1 participant failed at least one of the attention checks (Q3: Please select “Moderately” for this item and Q4: Please select “True” for this item in Phase 2 survey), leaving a final dataset of 117 valid responses. Among the 117 valid responses, 73.5% ($n = 86$) identified as having been victimized by at least one scam in the past (victimized group), while 26.5% ($n = 31$) reported never having been scammed (non-victimized group).

4. Results

As shown in Table 1, a total of 117 valid responses were collected. The sample included 70 female participants, 43 male participants, 2 individuals who preferred not to disclose their gender, and 2 individuals who identified as non-binary. Participants ranged in age from 20 to 77 years, with a mean age of 40.52 and a median age of 38.

The distribution of annual income and educational background is given in Table 1. Because age and income did not meet the assumption of normality, Mann–Whitney U tests were used to analyze these variables. For gender comparisons, a Chi-square test was conducted for only male and female categories ($\chi^2(1) = 3.2045, p = 0.0734$). A Fisher’s Exact Test was used when including the two non-binary participants ($p = 0.1416$). The statistical results (Table 1) were as follows: age ($U = 1536.0, p = 0.1408$), income ($U = 1304.0, p = 0.6382$), and education ($U = 1506.0, p = 0.2695$). For further analysis, a logistic regression model was conducted with age, gender, income, and education as predictors of victimization status, restricted to a complete-case sample ($n = 110$) in which all four predictors were observed with no missing values. The model showed that no variable to reach statistical significance after adjustment: age ($p = 0.1116$), income ($p = 0.3655$), education ($p = 0.5279$), and gender ($p = 0.0625$). This is consistent with the unadjusted analyses.

Based on these findings, we conclude that there were no significant differences in age, income, education level, or gender between the victimized and non-victimized groups.

Table 1
Participant demographics (N = 117).

Category	Details	Victimized	Non-victimized
Gender			
Female	59.83% (70)	68.57% (48)	31.43% (22)
Male	36.75% (43)	83.72% (36)	16.28% (7)
Non-binary	1.71% (2)	50.00% (1)	50.00% (1)
Prefer not to disclose	1.71% (2)	50.00% (1)	50.00% (1)
Age range			
	20–77		
Education level			
Less than high school:	1.71% (2)	50.00% (1)	50.00% (1)
High school/GED	15.38% (18)	61.11% (11)	38.89% (7)
Some college:	21.37% (25)	76.00% (19)	24.00% (6)
2-year degree	11.11% (13)	92.31% (12)	7.69% (1)
4-year degree	37.61% (44)	65.91% (29)	34.09% (15)
Master's degree	11.11% (13)	100.00% (13)	0.00% (0)
Doctoral degree	1.71% (2)	50.00% (1)	50.00% (1)
Annual income			
\$0	3.42% (4)	75.00% (3)	25.00% (1)
\$1–\$9999	13.68% (16)	56.25% (9)	43.75% (7)
\$10,000–\$24,999	16.24% (19)	73.68% (14)	26.32% (5)
\$25,000–\$49,999	26.50% (31)	87.10% (27)	12.90% (4)
\$50,000–\$74,999	17.95% (21)	76.19% (16)	23.81% (5)
\$75,000–\$99,999	7.69% (9)	77.78% (7)	22.22% (2)
\$100,000–\$149,999	7.69% (9)	66.67% (6)	33.33% (3)
\$150,000 or more	4.27% (5)	60.00% (3)	40.00% (2)
Prefer not to answer	2.56% (3)	33.33% (1)	66.67% (2)
Statistical tests			
Age (Mann–Whitney U)	$U = 1536.0, p = 0.1408$		
Income (Mann–Whitney U)	$U = 1304.0, p = 0.6382$		
Education (Mann–Whitney U)	$U = 1506.0, p = 0.2695$		
Gender (Chi-square; binary only)	$\chi^2(1) = 3.2045, p = 0.0734$		
Gender (Fisher's Exact Test; incl. non-binary)	$p = 0.1416$		
Logistic regression			
Age	OR = 1.0316, 95% CI = [0.9928, 1.0718], $p = 0.1116$		
Income	OR = 0.8671, 95% CI = [0.6366, 1.1810], $p = 0.3655$		
Education	OR = 1.1189, 95% CI = [0.7893, 1.5862], $p = 0.5279$		
Gender [T.Male] (binary only)	OR = 2.6858, 95% CI = [0.9497, 7.5960], $p = 0.0625$		

Table 2

Association between each test accuracy and victimization status, assessed using both chi-square tests and logistic regression adjusting for four demographic control variables (age, gender, income, and education). Only items reaching statistical significance ($p < 0.05$) are shown; the complete set of comparisons across all 12 test scenarios is provided in [Appendix \(Tables 8, 9\)](#). The asterisk (*) indicates statistical significance at $p < 0.05$.

Category	Test type	Analysis	Test No.	Victimized		Non-victimized		n	Statistic	p-value
				Incorrect	Correct	Incorrect	Correct			
Fraudulent	Phone-Based	Chi-square	2	49 (67%)	24 (33%)	10 (38%)	16 (62%)	99	$\chi^2(1) = 6.5407$	0.0105*
			6	24 (30%)	57 (70%)	2 (7%)	27 (93%)	110	$\chi^2(1) = 6.1142$	0.0134*
		Logistic	2	48 (69%)	22 (31%)	8 (36%)	14 (64%)	92	OR = 0.25 [0.08, 0.71]	0.0098*
			6	23 (29%)	55 (71%)	2 (8%)	23 (92%)	103	OR = 0.21 [0.04, 1.00]	0.0496*

4.1. Test results for scam scenarios (RQ1)

To address **RQ1 (Do participants with scam victimization experience exhibit greater accuracy in detecting scams compared to individuals with no history of victimization?)**, we conducted a comprehensive comparative analysis of response accuracy using the Chi-Square test and Logistic Regression, and accuracy scores using the Mann–Whitney U test between victimized and non-victimized participants across phone-based scam scenarios (Tests 2, 4, and 6) and email-based scam scenarios (Tests 7, 10, and 12). Results revealed that victimized participants performed significantly worse in phone-based scam scenarios, whereas no significant differences emerged in email-based scenarios, suggesting modality-specific vulnerabilities among scam victims.

Response accuracy (Chi-Square tests). As shown in [Table 2](#), Chi-Square tests of independence revealed significant differences in response correctness between groups for two tests, Test 2 ($\chi^2(1) = 6.5407, p = 0.0105$) and Test 6 ($\chi^2(1) = 6.1142, p = 0.0134$), both phone-based scam scenarios. Victimized participants answered Test 2 correctly in only 33% of cases ($n = 24$), compared to 62% ($n = 16$) among non-victimized participants, representing a 29% gap. A similar pattern

emerged for Test 6, with 70% correct ($n = 57$) in the victimized group versus 93% ($n = 27$) in the non-victimized group, a 23% difference. The remaining ten tests showed no significant group differences (all $p > 0.05$).

Adjusted response analyses (Logistic Regression). To determine whether the observed associations persisted after accounting for demographic factors that might be associated with people's scam susceptibility, we conducted logistic regression analyses predicting victimization status from each test's correctness, controlling for age ([James et al., 2014](#); [Hancock et al., 2025](#); [Sirohi and Misra, 2024](#); [Deliema et al., 2020](#)), gender ([James et al., 2014](#); [Sirohi and Misra, 2024](#); [Deliema et al., 2020](#)), income ([Sirohi and Misra, 2024](#); [Beach et al., 2023](#); [Ahmad, 2025](#)), and education ([Hancock et al., 2025](#); [Beach et al., 2023](#); [Ahmad, 2025](#)) ([Table 2](#)). Participants who reported their gender as “prefer not to disclose” or “prefer to self-describe” were excluded due to insufficient cell sizes for regression analysis ($n = 4$). The significant effects observed in the Chi-square tests remained significant after demographic adjustment for Test 2 (OR = 0.25, 95% CI = [0.08, 0.71], $p = 0.0098$) and Test 6 (OR = 0.21, 95% CI = [0.04, 1.00], $p = 0.0496$), indicating that the association between phone-based scam identification and victimization

Table 3

Mann–Whitney U Test results for selected comparisons between victimized and non-victimized groups. Only items reaching statistical significance ($p < 0.05$) are shown; the complete set of comparisons across all test scenarios and response tendencies is provided in [Appendix \(Table 10\)](#). The asterisk (*) indicates statistical significance at $p < 0.05$.

Category	Test type	Test No.	Victimized	Non-victimized	Mann–Whitney U
Fraudulent	Phone-Based	Test 2	($n = 73$, $M = 2.19$, $SD = 1.12$)	($n = 26$, $M = 2.69$, $SD = 1.14$)	* ($U = 697.5$, $p = 0.0383$)
		Test 6	($n = 81$, $M = 3.21$, $SD = 1.43$)	($n = 29$, $M = 3.97$, $SD = 1.10$)	* ($U = 829.5$, $p = 0.0161$)
Fraudulent	Phone-Based	Test 2, 4, 6	($n = 86$, $M = 2.85$, $SD = 1.01$)	($n = 31$, $M = 3.36$, $SD = 0.86$)	* ($U = 909$, $p = 0.0086$)
False Negative	All types	All test cases	($n = 86$, $M = 1.49$, $SD = 1.26$)	($n = 31$, $M = 0.84$, $SD = 1.02$)	* ($U = 1746$, $p = 0.0081$)

status is not attributable to demographic confounding. No other test reached significance in the adjusted models.

Accuracy scores (Mann–Whitney U tests). Because the normality assumption was violated, we used Mann–Whitney U tests to compare accuracy scores between groups ([Table 3](#)). Statistically significant differences emerged for Test 2 ($U = 697.5$, $p = 0.0383$), Test 6 ($U = 829.5$, $p = 0.0161$), the composite Phone-Based Fraudulent score ($U = 909$, $p = 0.0086$), and False Negative responses ($U = 1746$, $p = 0.0081$). Post-hoc analysis revealed consistently lower accuracy scores for the victimized group: Test 2 (Victimized Mean = 2.19, Non-Victimized Mean = 2.69), Test 6 (Victimized Mean = 3.21, Non-Victimized Mean = 3.97), and Phone-Based Fraudulent Tests (Victimized Mean = 2.85, Non-Victimized Mean = 3.36).

These findings collectively suggest that the victimized group performs poorer than the non-victimized group on Phone-Based scenarios. Furthermore, the victimized group exhibited a tendency towards identifying fraudulent scenarios as legitimate. This pattern suggests that *Individuals who fell for scams perform worse in identifying phone-based scams compared to people who have not fallen for scams in the past* (**Finding 1**)

4.2. Test results for legitimate scenarios (RQ2)

To test RQ2 (Do participants with scam victimization experience exhibit greater accuracy in detecting legitimate communication compared to individuals with no history of victimization?), we compared response accuracy and accuracy scores between victimized and non-victimized participants on legitimate scenarios, Test 3 (phone-based) and Tests 8, 9, and 11 (email-based), using Chi-Square tests, Logistic Regression, and Mann–Whitney U tests. As shown in [Table 8](#), [9](#), and [10](#), across all four tests, the majority of participants in both groups correctly identified the legitimate scenarios, and no significant between-group differences in response accuracy were observed. Logistic regression analyses, adjusting for age, gender, income, and education, likewise yielded no significant effects on victimization status in any legitimate-scenario test. Mann–Whitney U tests on accuracy scores produced the same pattern that mean scores for the victimized and non-victimized groups were comparable across Test 3 (4.01 vs. 4.17), Test 8 (4.41 vs. 4.45), Test 9 (3.89 vs. 3.59), and Test 11 (3.87 vs. 3.90), with no significant differences. Thus, the results suggest that *individuals with a history of scam victimization demonstrated comparable accuracy in identifying legitimate communications via email and telephone compared with those without such victimization experiences*. (**Finding 2**)

4.3. Test results for ambiguous scenarios (RQ3)

To test RQ3 (Do participants with scam victimization experience differ while labeling ambiguous scenarios as scams (or legitimate) compared to individuals with no history of victimization?), we compared response patterns (scam vs. non-scam classification) and accuracy scores between victimized and non-victimized participants on ambiguous scenarios (Tests 1 and 5) using Chi-Square tests, Logistic Regression, and Mann–Whitney U tests. As shown in [Table 8](#), [9](#), and [10](#), Chi-Square analyses revealed no significant association between victimization status and scam/non-scam classification for either Test 1

($\chi^2(1) = 0.0001$, $p = 0.9903$) or Test 5 ($\chi^2(1) = 0.5237$, $p = 0.4693$), and distributions of “Scam” and “Non-Scam” responses were highly similar between groups. Logistic regression analyses adjusting for age, gender, income, and education confirmed these findings, with no significant effects on victimization status for either test (both $p > 0.05$). Mann–Whitney U tests on accuracy scores yielded the same pattern, and mean scores were comparable between the victimized and non-victimized groups for Test 1 (2.11 vs. 2.13) and Test 5 (2.68 vs. 2.62). Based on our findings regarding ambiguous scenarios, we conclude that *participants with a history of scam victimization exhibit suspicion levels comparable to those without such experiences*. (**Finding 3**)

4.4. Confidence & helpfulness ratings (RQ 4 & RQ 5)

To test RQ4 (Do participants with scam victimization experience exhibit increased confidence in detecting scams compared to individuals with no history of victimization?) and RQ5 (Do participants with scam victimization experience perceive their past experiences as beneficial while identifying scams compared to individuals with no history of victimization?), we computed an average perceived helpfulness score and an average confidence score across the 12 scenarios for each participant, then compared these scores between two groups using Welch’s *t*-tests and Logistic Regression adjusting for age, gender, income, and education. The results revealed no statistically significant differences between the two groups across all scenarios, suggesting that victimized individuals have comparable levels of self-efficacy and perceived usefulness of past experience to non-victimized participants ([Table 4](#)).

Based on our findings, we conclude that *the level of perceived confidence in detecting scams and the usefulness of past victimization experiences were not significantly different between the victimized and non-victimized groups* (**Finding 4 & 5**)

5. Qualitative analysis

Our quantitative findings suggest that the victimized group performed poorer than the non-victimized group in phone-based scenarios (Test 2 and 6) and demonstrated a comparable level of perceived confidence and usefulness of past experience. This study provides compelling evidence that prior experience with scam victimization does not automatically build sufficient resilience against future fraud attempts, especially in scenarios lacking clear technical indicators, such as phone-based scams. The findings suggest that individuals who have previously fallen for scams do not necessarily learn enough from these experiences to draw effective lessons. In fact, they seem to remain vulnerable to future scams, most likely due to a persistent trust bias and a decision-making style that relies heavily on past experiences and/or superficial cues. In contrast, the results from email-based scenarios revealed no significant differences between the two groups. This suggests that technical cues, such as email headers, sender addresses, and formatting, are accessible and useful to both groups. It implies that the ability to detect fraud in these contexts may be more reliant on identifiable markers. In addition, complementary to our findings, [DeLiema et al. \(2024\)](#) demonstrated that the time interval between victimization incidents decreased with increasing prior victimization. Although the underlying reasons are unclear, this suggests that initial exposure to scams may not

Table 4

Association between overall helpfulness score/confidence score and victimization status, assessed using both Welch's *t*-Test and logistic regression adjusting for four demographic control variables (age, gender, income, and education). The asterisk (*) indicates statistical significance at $p < 0.05$.

Test	Victimized	Non-victimized	Statistics	p-value
Helpfulness score (Welch's t-test)	(n = 86, M = 3.24, SD = 0.91)	(n = 31, M = 2.92, SD = 1.04)	$t(47.6) = 1.4825$	0.1448
Confidence score (Welch's t-test)	(n = 86, M = 3.74, SD = 0.67)	(n = 31, M = 3.44, SD = 0.85)	$t(44.1) = 1.7617$	0.0850
Helpfulness score (Logistic)	(n = 83, M = 3.24, SD = 0.92)	(n = 27, M = 3.01, SD = 1.04)	OR = 1.297 [0.798, 2.107]	0.2945
Confidence score (Logistic)	(n = 83, M = 3.74, SD = 0.68)	(n = 27, M = 3.56, SD = 0.83)	OR = 1.392 [0.726, 2.671]	0.3195

Table 5

The top codes (cues) identified by participants for correct decisions for each case. For ambiguous tests, the top codes (cues) reflect participants' reasoning for judging a test as a legitimate scenario.

Category	Test type	Test No.	Victimized (Correct)	Non-victimized (Correct)
Fraudulent	Phone-Based	Test 2	Suspicious Invoice (10), Language/Voice Anomalies (5), Use of Urgency Tactics (5)	Suspicious Invoice (5), Language/Voice Anomalies (4), Personal Data/Money Request (3), Had Similar Experience Before (3)
		Test 4	Suspicious Cues (e.g., "Secure Server", "Fake Text Message") (28), Language/Voice Anomalies (5), Use of Urgency Tactics (2), Lack of Verification (2)	Suspicious Cues (e.g., "Secure Server", "Fake Text Message") (14), Use of Urgency Tactics (3), Language/Voice Anomalies (2), Had Similar Experience Before (2)
		Test 6	Personal Data/Money Request (43), Unsolicited Contact (7), Suspicious Text Format (7)	Personal Data/Money Request (24), Unsolicited Contact (5), Lack of Verification (3)
	Email-Based	Test 7	Suspicious Financial Demand (54), Suspicious Email Address (24), Language/Format Anomalies (14)	Suspicious Financial Demand (18), Suspicious Email Address (10), Language/Format Anomalies (7)
		Test 10	Suspicious Email Address (29), Language/Format Anomalies (16), Suspicious Transaction Details (11)	Suspicious Email Address (14), Language/Format Anomalies (8), Suspicious Transaction Details (2)
		Test 12	Request to Click/Download (24), Suspicious Email Address (20), Urgency Tactics (16)	Request to Click/Download (11), Suspicious Email Address (7), Urgency Tactics (6), Language/Format Anomalies (6)
Legitimate	Phone-Based	Test 3	Family/Friend Relationship (31), No Personal Data/Money Request (24)	Family/Friend Relationship (16), No Personal Data/Money Request (10)
	Email-Based	Test 8	No Personal Data/Money Request (29), Verified Sender Identity (18), Informative Content (15)	No Personal Data/Money Request (12), Informative Content (9), Verified Sender Identify (8)
		Test 9	No Language/Format Anomalies (23), Legitimate Email Address (16), No Personal Data/Money Request (8)	No Language/Format Anomalies (10), Legitimate Email Address (3)
		Test 11	EDU Domain (22), Legitimate Password Reset Protocol (12), No Links (8)	EDU Domain (9), No Links (4), Instructions for Verification (3), Legitimate Password Reset Protocol (3)
Category	Test type	Test No.	Victimized (Marked as Legitimate)	Non-victimized (Marked as Legitimate)
Ambiguous	Phone-Based	Test 1	Internal Transfer (25), Customer-Initiated Contact (19), No Sensitive Info Shared (18)	Internal Transfer (11), No Sensitive Info Shared (11), Customer-Initiated Contact (9)
		Test 5	Legitimate Password Reset Protocol (13), Caller-Initiated Contact (9), Legitimate Phone Number Assumed (6)	Legitimate Phone Number Assumed (5), No Personal Data/Money Request (4), Legitimate Password Reset Protocol (4), Caller-Initiated Contact (3)

only fail to improve subsequent scam-detection capabilities relative to non-victimized individuals but also contribute to future victimization without appropriate intervention strategies. To better understand the underlying reasons, we conducted a detailed qualitative analysis of all test scenarios to identify the top reasons participants cited for their decisions. We present the analysis in two steps. First, we discuss the top reasons participants gave for making the correct decisions (Table 5). Next, we discuss the top reasons participants gave for making incorrect decisions (Table 6).

5.1. The underlying reasons behind correct identification

After thoroughly analyzing participants' comments explaining why they perceived each test as fraudulent or legitimate, we identified the top reasons influencing their correct decisions. These cues/codes, along with participants' reasoning for categorizing ambiguous tests as either scams or non-scams, are summarized in Table 5. To ensure more reliable and informative results, we included only the codes mentioned at least twice and excluded "Wrong Understanding/Unmatched Answer/Not Sure". Additionally, if the third and fourth codes were mentioned the same number of times, we listed both. "N/A" indicates cases where no codes were identified, either because the code was

mentioned fewer than twice or because there was no comment in that codebook.

As Table 5 shows, the commonly cited reasons for identifying test scenarios correctly are often the same between the victimized and non-victimized groups. In these scenarios, red flags such as *requests for personal data or money*, *unsolicited contact*, *use of urgency tactics*, *use of suspicious language*, *formatting issues*, or *sender details* were key indicators. Details are presented below.

5.1.1. Presence of suspicious cues

For multiple cases, participants pointed out the presence of suspicious cues as the reason for their decisions. For example, in Test 2, participants flagged the PayPal vishing scam due to the suspicious invoice and voice anomalies (*"Call sounded suspicious when asked for invoice#"*, *"I think this might be a scam,I think the customer service number on the false invoice caused the customer to call an entity that is not PayPal"*, *"Immediately saying that someone made an account in her name even though it sounds like she has an account already made for PayPal"*, *"The automated woman sounds like she is trying to scam someone. Her English was not correct, and the way she raised her voice at the end sounds like a call to action to upset the caller and make her give her information."*).

Table 6

The top codes (cues) identified by participants for incorrect decisions for each case. For ambiguous tests, the top codes (cues) reflect participants' reasoning for judging a test as a scam scenario.

Category	Test type	Test No.	Victimized (Incorrect)	Non-victimized (Incorrect)
Fraudulent	Phone-Based	Test 2	Legitimate Customer Service (20), Caller-Initiated Contact (17), Invoice Number Provided (14)	Legitimate Customer Service (6), Invoice Number Provided (4), Caller-Initiated Contact (3)
		Test 4	Victimization by Fraud (Not the phone call) (9), Caller-Initiated Contact (8), Authentic Banking Interaction (4), Caller's Credibility (4)	Authentic Banking Interaction (4), Credible Secure Server (2)
		Test 6	Legitimate Bank Security Protocol (12), Bank-Initiated Alert (8), Had Similar Experience Before (5)	N/A
	Email-Based	Test 7	N/A	N/A
		Test 10	Legitimate Receipt/Invoice Number (12), Legitimate Contact Number (2), No Personal Data/Money Request (2)	Legitimate Receipt/Invoice Number (4)
		Test 12	No Personal Data/Money Request (4), Legitimate Email Address (3), No Language/Format Anomalies (2)	N/A
Legitimate	Phone-Based	Test 3	Unclear Context/Purpose (11), Unnatural/Suspicious Flow (8), Potential AI-Duplicated Voice (3)	Potential AI-Duplicated Voice (3), Unclear Context/Purpose (3), Unnatural/Suspicious Flow (2)
		Test 8	Suspicious Links (4), Unsolicited Contact (2)	Suspicious links (2)
	Email-Based	Test 9	Suspicious Email Address (10), Suspicious Links (10), Language/Format Anomalies (6)	Suspicious Email Address (7), Suspicious Links (3)
		Test 11	Personal Data/Money Request (10), Unfamiliar Conduct (6), Language/Format Anomalies (3)	Personal Data/Money Request (3)
Category	Test type	Test No.	Victimized (Marked as Scam)	Non-victimized (Marked as Scam)
Ambiguous	Phone-Based	Test 1	Lack of Verification (14), Potential AI-Duplicated Voice (3), Unverified Phone Number (3)	Lack of Verification (6), Potential AI-Duplicated Voice (2), Suspicious Transfer Practices (2)
		Test 5	Account/Phone Takeover Risk (12), Suspicious Password Reset Protocol (11), Suspicious Links (11)	Suspicious Password Reset Protocol (7), Account/Phone Takeover Risk (4), Lack of Verification(3), Language/Voice Anomalies (3)

5.1.2. Use of urgency tactics

Many correctly identified the use of urgency to pressure participants. For instance, in Test 4, the Wells Fargo impersonation was widely recognized as fraudulent because of suspicious requests to access a "secure server" under the pretense of urgency (*"He's saying that he signed an agreement is a red flag and then the bank informing him to be transferred to a secure server"*, *"Telling him to call in to Wells Fargo and then getting him on the hook for the 'secure server', 'This call played on urgency and tried to get them to connect to a different server.'"*).

5.1.3. Lack of use of proper verification protocol

In case of Test 6, the non-victimized group correctly identified "Lack of Verification" as a clue (*"It's unclear of the source of the text and call. It could be gathering identity theft information and not be the bank."*, *"Scammy looking text and provided no indication that it was in fact the bank the guy was talking to..."*, *"Anyone can send a text like that,, It could be anyone behind that number."*). Interestingly, the victimized group did not identify a lack of proper verification as a top reason.

5.1.4. Presence of immediate demand

Many participants cited immediate demands as a reason to flag a communication as a scam. For instance, they correctly noted the suspicious financial demand in case of Test 7 (*"The email sounds fishy. also it's asking someone to send money"*, *"You never have to pay to receive money!"*, *"Financial request seems very suspicious"*) or a request to download software in case of the Cisco VPN scam (Test 12) (*"the link can be phishing"*, *"Never click on a link that is sent to you through an email"*, *"It's asking them to download an unknown source."*). These cases demonstrated participants' ability to detect common social engineering strategies, though occasional misinterpretations showed how seemingly legitimate details can be used to exploit trust.

5.1.5. Legitimate scenarios

In the case of a legitimate scenario (Test 3), it was mostly recognized as legitimate due to the established relationship and lack of sensitive requests (*"It's two friends having a conversation"*, *"Sounds like friend's talking to one another about life"*, *"There was no discussion about money....."*, *"No personal information was collected, this seems like a call between friends."*), but a few of the participants worried about unclear context and that it might be an AI voice duplicated by a scammer in order to build trust before scamming. Tests 8, 9, and 11 were generally identified correctly. Test 8 was recognized as legitimate due to its informative nature, verified sender identity, and lack of requests for sensitive information, and Test 9 was identified as legitimate mainly because of no language/format anomalies shown in the email, use of legitimate email address, and no personal data/money request in the content (*"The ad was not asking for information that was personal. Looked legit to me."*, *"The email came from the hawks email, and wasn't really sounding scammy"*, *"email looks like it is from the legit team. Logo and letterhead on point."*). Similarly, Test 11 was believed legitimate because of its use of an EDU domain and adherence to the proper password reset protocol (*"This is from a .edu site and unlikely to be a scam....."*, *"the password recovery looks familiar and the most usual way of resetting a password"*, *"It gives detailed information & who to contact at company."*).

5.1.6. Ambiguous scenarios

In the case of ambiguous test scenarios, both groups of participants cited similar reasons, such as internal transfer between accounts owned by the same owner, customer-initiated contact, and the lack of sharing of sensitive information (Test 1). In Test 5, the commonly cited reasons were the use of a legitimate password reset protocol, caller-initiated contact, and the use of a legitimate phone number (assumed by some participants). The non-victimized group also mentioned the lack of a request for personal data/money as a clue, though this does not necessarily imply a legitimate scenario in real life.

5.2. The underlying reasons behind erroneous identification

In case of incorrect reasoning, as shown in Table 6, there are some notable differences in the incorrectly identified cues between the two groups. The key distinction lies in the victimized group's greater reliance on surface-level indicators of legitimacy and an over-reliance on familiarity. The key findings are discussed below.

5.2.1. Perceived credibility of the caller and the message

Cues such as "Caller's Credibility" (trusted by participants) appear only in the victimized group's incorrect responses. Specifically, a central theme across responses is the acceptance of legitimacy simply because the caller initiated contact to report a problem (*"The person was calling to report a scam."*, *"The person is reporting a scam, not attempting to scam anyone."* (Test 2)). Many participants assumed the interaction was legitimate simply because the caller reached out to report a problem, revealing a key misunderstanding of how phishing scams typically work: scammers often pose as helpful agents. This initial framing seemed to lower participants' guard. In case of Test 2, many participants judged the legitimacy of the call based on whether the caller, rather than the organization, initiated it (*"The caller called PayPal directly about a charge on her account that she did not make..."*, *"The woman called paypal herself to report an issue rather than them calling her..."*). A similar pattern emerged in Test 4, where a significant number of participants equated caller-initiated contact with genuine communication (*"The man called the bank"*, *"I don't think the phone call itself is a scam, because he called the company..."*). Participants appear to miss the point that the victim initiated the call in response to scam messages in the first place.

Although indicators of trust-based heuristics, such as "Authentic Banking Interaction", appear for both groups, "Credible Secure Server" appears only for the non-victimized group. This suggests a strong tendency among some of these participants to rely on perceived credibility when deciding whether a communication is legitimate. Several respondents in Test 6 focused on the caller's perceived attempts to verify. The fact that the call originated from the same number as the text message and that the agent attempted to verify identity were interpreted as positive signs (*"The number the agent called was the same one the text was sent to, plus the agent made an attempt to verify that she was speaking to the correct person."*, *"The account owner had to give his correct details (DOB) before the caller from the bank told him what had happened."*). Simple "verification" steps, such as matching the caller's number or answering a few generic questions, were often mistaken for proof that the call was legitimate. This overlooks how easily scammers can spoof phone numbers and/or use superficial verification questions. The comments demonstrate a misunderstanding of what constitutes genuine security measures.

5.2.2. Reliance on familiarity

Participants were often noted to rely on familiarity with similar past experiences while determining the legitimacy of an interaction. For instance, in Test 6, participants frequently cited their own experiences receiving similar alerts from financial institutions (*"I do not believe this is a scam because I have received similar text messages from my financial institution regarding suspicious charges."*, *"The bank is verifying the info from the caller."*).

This reliance on their past experiences with similar, legitimate interactions to assess a situation led to a false sense of security, as reflected in the top three cues: "Legitimate Bank Security Protocol", "Bank-Initiated Alert", and "Had Similar Experience Before". These findings indicate that familiarity with past interactions and excessive trust in others may have made the victimized group more susceptible to scams in certain contexts.

5.2.3. Presence of superficial cues

Some participants focused on the presence of superficial cues such as invoice numbers or receipts as strong evidence of legitimacy. For instance, in Test 2, participants mentioned reasons such as (*"The woman asked for invoice number to check what was the bill for..."* and *"I think the lady is a legit customer who owns the Paypal account or else she would not have had a matching invoice number the woman from PayPal could look up."*). Similarly, in Test 10, some noted (*"This seems to be a legitimate email from PayPal of a receipt for payment on a TV..."*, *"The text sounded appropriate, and there were a lot of account numbers and transaction numbers posted."*). These reasons showed that participants viewed an invoice number or a receipt as strong proof of authenticity, a critical misjudgment, as these can be easily fabricated.

5.2.4. Absence of immediate demand

Some victimized participants focused on the absence of immediate demands, such as "No Personal Data/Money Request". For instance, in Test 2, a significant portion of responses cite the absence of explicit requests for sensitive personal information as justification (*"No personal information was shared between the woman and PayPal"*, and *"No personal ID given"*). Similarly, for Test 6, 10, and 12, participants' responses emphasized the lack of immediate financial loss as an indicator of legitimacy (*"There was no loss"* (Test 6), *"The card number was not discussed on the phone."* (Test 6), *"It's not really asking me for money"* (Test 10), and *"Just updating system"* (Test 12)).

5.2.5. Legitimate scenarios

For the legitimate test scenarios (Test 3 (Phone-based) and Tests 8, 9, and 11 (Email-based)), participants were noted to rely on cues such as "Suspicious Links", "Unsolicited Contact", "Suspicious Email Addresses", and "Language/Format Anomalies". In Test 3, some were concerned that a duplicated AI voice was being used. Importantly, this demonstrates the difficulty of identifying scams based solely on superficial technical cues.

5.2.6. Ambiguous scenarios

For the ambiguous tests, in the case of Test 1, some participants were concerned about insufficient verification, use of an unverified phone number, and the possibility of a duplicated AI voice. Similarly, in Test 5, participants suspected fraud due to concerns about phone-takeover risk, lack of verification, language anomalies, and suspicious links.

Our analysis suggests that victimized and non-victimized groups often use similar reasoning when evaluating the legitimacy of different scenarios and often rely on trust-based heuristics (e.g., familiarity with past experiences, credibility of callers, overconfidence in the authenticity of the communication protocol/server) rather than objective measures.

6. Discussion

Our quantitative findings shed light on differences between victimized and non-victimized groups in their ability to detect scams, perceived confidence, and perceived helpfulness of past experience. Our study expands upon prior research (DeLiema et al., 2024, 2025b; Whitty, 2019; DeLiema et al., 2023; Chen et al., 2020) by examining participants' past scam victimization status (including both financial and data loss as forms of victimization) and by evaluating their performance using three types of scenarios (i.e., legitimate, fraudulent, and ambiguous) across two different modalities (i.e., email-based and phone-based). The results suggest that participants who had previously been victimized performed significantly worse at identifying fraudulent phone-based scenarios than non-victims, while showing no significant differences in identifying legitimate or ambiguous scenarios. To understand the underlying factors behind susceptibility to scams,

we conducted a qualitative analysis. The findings indicate that scam susceptibility is often not due to a lack of knowledge about technical indicators, but rather to specific cognitive biases, such as reliance on trust, focusing on immediate losses, and over-reliance on similar experiences. The implications of our findings are discussed below.

6.1. The lack of effect of demographic variables

Many studies have shown that demographic variables are significant predictors of scam victimization. In prior efforts, younger adults are noted to be involved with scams like online shopping and basic online scams at a higher frequency, whereas older adults are associated with scams that involve romance, technical support, investment, and mail scams, with repeat victimization rising sharply for ages in the 70s and 80s (DeLiema and Witt, 2024; Daengsi et al., 2022; Hanoch and Wood, 2021; DeLiema et al., 2024; Deliema et al., 2020). In addition, some studies have shown that gender also plays an important role in terms of scam victimization. Specifically, men are more likely to be scammed by opportunity and investment fraud, while women are more likely to be scammed by consumer scams (Whitty, 2020; Lokanan and Liu, 2021; Sheng et al., 2010). Furthermore, education and income level have been shown to have dual effects. Communities with more education have shown a lower overall fraud rate, yet individually, higher education, financial literacy, and asset accumulation can increase exposure to complex and persuasive investment scams due to overconfidence in financial literacy (Raval, 2021; Mao and Liu, 2023; Deliema et al., 2020; Lokanan and Liu, 2021). These demographic factors shape susceptibility to scams through digital exposure, technical skill gaps, financial engagement intensity, and overconfidence, rather than through fixed vulnerability. However, in our study, the absence of significant demographic differences is noteworthy. It underscores that anyone, regardless of age, education, or socioeconomic status, can fall victim to scams (Hanoch and Wood, 2021). This finding emphasizes the psychological and manipulative nature of scams, which exploit universal cognitive vulnerabilities (Coluccia et al., 2020; Whitty, 2019; Deliema et al., 2020) rather than targeting specific demographic groups. It also highlights the limitations of prevention strategies that rely solely on demographic profiling. Instead, interventions should focus on identifying and addressing cognitive vulnerabilities and behavioral patterns.

6.2. The tendency to rely on trust-based heuristics

A closer look at qualitative data revealed that participants who had previously been victimized often relied on superficial cues (e.g., invoice numbers, familiar formatting, or the mere fact that someone proactively contacted them). This behavior reflects a trust bias, where individuals are inclined to accept narratives that feel familiar or align with their expectations, even when warning signs are present. The trust bias manifests in different forms. For instance, many of these participants attributed benevolent intent to the caller, which illustrates their tendency to interpret interactions through a lens of misplaced trust. Participants seem to reason that someone who proactively reports fraud could not be engaging in fraudulent activity, demonstrating a fundamental misunderstanding of vishing tactics. This demonstrates an inherent trust bias, particularly strong in those who have previously experienced fraud and may be seeking reassurance or resolution.

We also noted that many participants quickly assumed the interaction was genuine because it followed a familiar pattern: first a text message, then a call from someone claiming to be a bank representative. This sense of familiarity seemed to override deeper scrutiny, with participants relying more on how the interaction looked than on what was actually being said or done. This illustrates a dangerous reliance on familiar communication structures without critically evaluating the content, verifying the source, or assessing the verification processes used in the interactions. Their trust was shaped by past experiences

with real fraud alerts, which made them more likely to believe anything that resembled those experiences. It is crucial to note that banks rarely initiate contact in this manner, particularly requesting full personally identifiable information (PII) following unsolicited text messages. The participants seem to be operating under the assumption that if it looks like a standard fraud alert, it must be one, which may reflect a cognitive shortcut that bypasses critical thinking.

Interestingly, there were no significant differences in performance in identifying legitimate communications or responding to ambiguous tests. This suggests that the issue is not a general difficulty with information processing or discernment. Rather, the vulnerability seems specific to contexts in which trust is exploited and fraudulent messages mimic legitimate communication patterns. Victimized individuals were just as capable of recognizing genuine messages as their non-victimized peers, indicating that their overall suspicion level was not universally lower. Their susceptibility appears more targeted and context-dependent. Overall, this suggests that previous negative experiences are not being effectively translated into improved self-efficacy, possibly due to insufficient reflection and learning after a scam incident, and/or difficulty in applying lessons from one context to another.

6.3. The risk of scam detection based on superficial “signs”

A large number of responses across multiple test cases highlight a reliance on superficial signs, such as caller-initiated contact, an invoice number/receipt, and the use of a secure server, as proof of authenticity. While invoices and receipts are a normal part of genuine transactions, participants seemed to overlook that such details can be easily fabricated, and that an invoice number or a receipt does not make a call or an email trustworthy. This suggests that people may have a trust bias, a tendency to benevolently interpret certain cues (e.g., invoice numbers or receipts). In addition, a small group of participants accepted the scammer’s claim of transferring to a secure server as genuine, leading them to believe Test 4 represented a legitimate scenario (**“Sounded like a normal call and caller was being transferred to a secure connection”, “This sounded like a legit bank rep. She also mentioned connecting the customer to their secured portal to get his refund.”, and “Would connect him to the Wells Fargo secure server.”**). This finding indicates that some individuals not only readily trust certain cues but also tend to believe what they are told by someone without skepticism.

6.4. Focusing too much on “immediate” loss

In this study, we defined immediate losses as direct and readily recognizable harms, such as monetary loss or the disclosure of sensitive information (e.g., Social Security Number). Long-term losses, in contrast, are more diffuse and can unfold over time. For instance, scammers may request information that appears innocuous in isolation, such as a name, age, or gender, but that can be aggregated to map a victim’s social network and later used to defraud the victim’s family, friends, or the victim themselves through more targeted attacks. Recently, attackers have been observed to invest substantial time in building trust and emotional attachment before any exploitation occurs, so the eventual loss is temporally decoupled from the early interactions.

We noted that participants often attempt to assess the legitimacy of a scenario by whether the caller is asking for anything of value. Participants seem to believe that if no money was lost during the interaction, it could not be fraudulent, and they ignore the possibility of identity theft or future financial losses resulting from the compromise of sensitive information. In addition, this suggests people naturally tend to trust others who pretend to offer help. This shows a limited understanding of how scammers often operate gradually, building trust before extracting personal data, or using seemingly innocuous data to access or build the target’s profile. The focus on immediate loss obscures the long-term risks associated with compromising sensitive information. In addition, Test 11 showed that participants focused heavily on

cues indicating personal information requests in unfamiliar situations (e.g., the university password recovery confirmation protocol). Test 11 scenario did not actually request any personal details, however, the participants believed it was a scam because the email involved words such as password, email address, and phone number, which created an unsafe context and triggered risk signals for them (*“Sounds off to request such information”, “I always assume anything regarding your accounts and passwords is a scam”*). These findings suggest that people tend to focus on immediate losses rather than long-term losses or gains. Importantly, this tendency becomes more pronounced when people operate in an uncertain or unfamiliar environment where the type of message/communication is new to them, and they have no clear point of reference for judging its legitimacy. For instance, recipients who had never previously gone through a password recovery process were more susceptible to such emails, and recipients with limited prior engagement with cryptocurrency were more susceptible to emails promoting cryptocurrency. Thus, this can cause them to ignore legitimate communication as well.

6.5. Recommendations for scam prevention training

Our findings indicate that fraud prevention efforts must extend beyond technical awareness. While educating individuals about observable indicators, such as suspicious URLs, grammatical errors, and unverified sender addresses (Jampen et al., 2020; Weaver et al., 2021; Kumaraguru et al., 2009), is necessary, it is insufficient and may foster a false sense of security. We argue that effective scam prevention requires addressing cognitive biases such as trust bias, focus on immediate loss, and overreliance on similar experiences, in addition to those identified in previous studies (Waldman, 2020; Chen et al., 2025; Maulidi, 2020; Judges et al., 2017). Scam prevention training should therefore prioritize cultivating critical thinking, encouraging healthy skepticism, and equipping individuals to recognize manipulative tactics (Nguyen et al., 2023; Jensen et al., 2017; Vishwanath et al., 2018). Our recommendations are presented below.

6.5.1. Promoting verification by default

Our qualitative data show that a strong trust bias persists, especially over the phone-based scenarios. It suggests that richer and more personal communication channels (e.g., calls or chat) can make messages feel more credible and urgent, reducing the likelihood that people slow down and scrutinize what is being asked (Norris et al., 2019; Anderson et al., 2024). Email, by contrast, often contains more visible technical cues (e.g., sender address, URLs) and gives people more time and space to check details (Norris et al., 2019; Koning et al., 2024). A practical response could be to introduce *“verification by default”* as a standard behavior, such as “Hang up and call back using the number on the official website, your card, or a known statement”, to shift the task of verification away from in-the-moment judgment and onto a repeatable action. This kind of script is in line with recommendations to build friction and protective “defaults” into risky interactions so that users do not have to rely solely on self-control or perfect attention every time (Norris et al., 2019; Koning et al., 2024). The idea is less about making people suspicious of everything and more about making the safe choice.

6.5.2. Training focusing on interrogation

Our qualitative analysis demonstrates that victimized groups tend to focus more on immediate losses. A likely reason is that the “lesson” people extract from their victimization experience is narrow. Instead of understanding how the scam worked (i.e., how urgency, authority, or fake legitimacy were used), they often focus on the specific outcome: “I lost money that time, so next time I’ll be wary of money requests”, which leaves many blind spots. This is worrying because much of the real damage from modern scams occurs later, through identity theft (e.g., the creation of new accounts using the victim’s name, fraudulent

tax filings, and online account takeovers) (Koning et al., 2024; Anderson et al., 2024). To address this, training needs to explicitly unpack the process of a scam. Rather than simply saying “do not trust calls about your account”, a better approach is to help people ask “Why would a legitimate organization contact me this way?” and “What could they gain from this interaction?” This aligns with models showing that suspicion and more systematic thinking are key to resisting phishing, and that interactive practice can strengthen those processes without making people generally distrustful (Vishwanath et al., 2018; DeLiema et al., 2025b). The goal is to shift from “don’t trust XYZ” to “interrogate what’s happening here.” We therefore recommend **risk-framing training** that helps people shift from an immediate “cash loss” lens to a more realistic “identity and account risk” lens. One approach could be to walk through scenarios in which a person shares a Social Security Number and other PII in response to a request from a fraudulent organization, without any immediate loss/consequence. But later, that information is used to open credit lines or gain access to other services. In such a scenario, we may ask a person to reflect on “How did the organization contact the person?” “What did the organization gain from this interaction?” “How and when did the organization use the gained information fraudulently?” This kind of reframing can show that people often underestimate future risks and overweight visible, short-term outcomes (Ahmad, 2025), leading them to reflect on what truly happened there. Making the delayed consequences concrete and salient can help counteract the danger of focusing too much on immediate losses.

6.5.3. Development of post-incident reflection tools

Our results align with studies showing that high awareness, self-reported literacy, or frequent visits to safety websites often do not reduce fraud risk and can even be more common among those who have already been targeted or victimized (Whitty, 2019; Zhang and Ye, 2022; Balakrishnan et al., 2025). Experience and awareness without guided reflection can remain abstract and unconnected to real improvement. We recommend future research on the development of **post-incident reflection tools** that use short, structured prompts for people to complete after a scam attempt or a close call. These might ask: *What exactly happened? What signals or cues were present? Which cues did you notice? Which cues did you miss? What would you do differently next time?* This is conceptually similar to psychosocial interventions that aim to improve well-being and reduce loneliness, which have been associated with lower scam susceptibility over time (Sur et al., 2021; DeLiema et al., 2025a). Structured reflection will help transform a distressing experience into a learning episode rather than simply being remembered as “a bad event”.

6.6. Limitations of the study

We took several measures, such as screening and employing attention and manipulation checks, to ensure the study’s internal validity and data quality. However, the reported findings should be interpreted with the following limitations in mind.

First, we recruited participants through the Prolific platform who are adults residing in the United States. These participants possess the technical ability to use Prolific and complete online surveys, which may not be representative of the general population of the United States or of populations outside the United States.

Second, we defined victims as those who have lost money or other valuables (e.g., personal data) due to past scams. However, different types of past scam experiences may lead to different psychological and performance outcomes. For example, being scammed five times may have a different impact than being scammed once. Similarly, the consequences of losing \$10,000 are likely to differ from those of losing \$100, and scams involving the loss of sensitive personal data may carry different implications than those involving non-sensitive information. In addition, being victimized by a particular type of scam might affect

victims differently in their performance on the same and different types of scams in the future, and it is unclear whether the effects of one type of scam victimization generalize to other types. Future research should explore these nuances more deeply.

Third, in our study, participants were aware that they were taking part in simulated scam tests rather than confronting genuine, unsolicited scam attempts. This foreknowledge likely altered their mental state, vigilance level, and decision-making strategies compared with individuals who encounter scams in real-world contexts. Thus, the difference may limit the validity of some findings. The participants may have been more analytical and less emotionally reactive than typical individuals in response to an unanticipated scam. We plan to address this limitation in future work by designing more natural, contextually embedded simulations that approximate real-world exposure. In addition, we assessed participants' in-the-moment ability to detect scams, but did not conduct a longitudinal study. As a result, these findings may not generalize to participants' future performance or their ability to detect scams over time. Additional long-term research would be needed to determine how stable or transferable these skills are in real-world contexts.

Fourth, in our study, we currently exclude individuals who have received prior cybersecurity training to avoid potential confounding effects. Future research focusing on the effect of prior cybersecurity training on scam susceptibility may include participants and categorize them by the type of training they completed, such as by its format, content, depth, and recency. This would allow us to compare the effectiveness of different training approaches and assess how trained individuals differ from those with no training and from those with other forms of training. By examining these groups side by side, we can gain a clearer understanding of which types of training offer meaningful protection and whether certain programs outperform others, which was not the focus of our study.

Fifth, while scam, legitimate, and ambiguous tests were deliberately interleaved to reduce cumulative order effects, we did not employ counterbalancing (e.g., a Latin Square design) across participants. Participants' awareness of fraud-related content may have increased as they progressed through the test, potentially influencing responses on later items. However, as any such effect would apply equally to the victimized and non-victimized groups, it is unlikely to account for the between-group differences we report. Further, the scam and legitimate scenarios vary in terms of the approaches and modality, which minimizes any such possible learning effect. Nonetheless, readers should interpret test-level comparisons with this consideration in mind, as the absolute accuracy level on later tests may reflect accumulated awareness.

Finally, our test scenarios did not include all types of scams, such as romantic, employment, home repair, or face-to-face physical scams. Future research should incorporate a broader range of scam types when designing test scenarios to more thoroughly assess participants' scam-detection abilities. In addition, our email-based scenarios were static rather than dynamic, meaning participants were unable to interact with the emails as they would in real-life situations (e.g., hovering over or clicking links). This static format may limit participants' ability to gather the critical information needed to make their decisions.

7. Conclusion

This study examined differences in scam detection between individuals who had previously fallen victim to scams and those who had not. A demographic analysis found no significant differences in age, gender, income, or education levels between the two groups. These findings suggest that demographic factors alone may not predict vulnerability to scams. However, the results showed that participants with prior victimization performed worse at identifying fraudulent phone-based communications. In contrast, both groups showed similar accuracy in recognizing legitimate communications and displayed similar levels of

suspicion in response to ambiguous scenarios. Further, the victimized group did not report greater perceived confidence or usefulness of past victimization experiences compared to the non-victimized group. Our qualitative analysis revealed several cognitive biases that lead participants to fail to correctly identify scams, including reliance on superficial cues (e.g., invoice numbers/receipts), familiarity with messaging formats, overtrusting caller-initiated contact, and past similar experiences. Their trust in familiar experience, proactive fraud alerts, and authority figures, such as bank employees, also contributed to misidentifying fraudulent situations. This pattern suggests that prior victimization does not serve as a protective learning experience. Instead, familiarity and reliance on trust-based heuristics can reduce skepticism. This research highlights the need for targeted interventions that go beyond raising basic awareness of common scams and technical indicators. Effective prevention strategies should focus on addressing cognitive biases and on creating post-incident reflection tools to turn a victimization experience into a meaningful lesson.

CRedit authorship contribution statement

Zhelun Rong: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Maifi Hasan Khan:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical approval

The study was approved by the BRANY IRB on January 22, 2025, under protocol number 24-380-910.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT to assist in improving readability and grammatical clarity. All AI-generated suggestions were carefully reviewed and, where appropriate, revised by the authors. The authors take full responsibility for the final content of the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

A.1. Supplementary tables

See [Tables 7–10](#).

Table 7

All categorized codes identified by the authors for each test scenario based on the correctness of participants' responses (Correct vs. Incorrect), regardless of their victimization status.

Test No.	Incorrect	Correct
Test 2	Invoice Number Provided, Legitimate Customer Service, Caller-Initiated Contact, No Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Suspicious Invoice, Language/Voice Anomalies, Use of Urgency Tactics, Personal Data/Money Request, Lack of Verification, Unverified Phone Number, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 4	Victimization by Fraud (Not the phone call), Caller-Initiated Contact, Authentic Banking Interaction, Caller's Credibility, Credible Secure Server, No Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Suspicious Cues (e.g., "Secure Server", "Fake Text Message"), Language/Voice Anomalies, Use of Urgency Tactics, Lack of Verification, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 6	Legitimate Bank Security Protocol, Bank-Initiated Alert, Caller-initiated Contact, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Personal Data/Money Request, Unsolicited Contact, Suspicious Text Format, Lack of Verification, Suspicious Transaction, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 7	Legitimate Language/Format, Legitimate Upgrade Protocol, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Suspicious Financial Demand, Suspicious Email Address, Language/Format Anomalies, Lack of Official Contact Info, Had similar experience before, Wrong Understanding/Unmatched Answer/Not sure
Test 10	Legitimate Receipt/Invoice Number, Legitimate Contact Number, No Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Suspicious Email Address, Language/Format Anomalies, Suspicious Transaction Details, Suspicious Phone Number, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 12	No Personal Data/Money Request, Legitimate Email Address, No Language/Format Anomalies, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Request to Click/Download, Suspicious Email Address, Use of Urgency Tactics, Language/Format Anomalies, Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 3	Unclear Context/Purpose, Unnatural/Suspicious Flow, Potential AI-Duplicated Voice, Personal Info Disclosure, Use of Urgency Tactics, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Family/Friend Relationship, No Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 8	Suspicious Links, Unsolicited Contact, Suspicious Source, No Personal Data/Money Request, Language/Format Anomalies, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	No Personal Data/Money Request, Verified Sender Identity, Informative Content, Legitimate Language/Format, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 9	Suspicious Email Address, Suspicious Links, Language/Format Anomalies, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	No Language/Format Anomalies, Legitimate Email Address, No Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 11	Personal Data/Money Request, Unfamiliar Conduct, Language/Format Anomalies, Suspicious Links, Suspicious Email Address, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	EDU Domain, Legitimate Password Reset Protocol, No Links, Instruction for Verification, Legitimate Language/Format, No Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
	Marked as Scam	Marked as Legitimate
Test 1	Lack of Verification, Potential AI-Duplicated Voice, Unverified Phone Number, Suspicious Transfer Practices, No Personal Data/Money Request, Language/Voice Anomalies, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Internal Transfer, Customer-Initiated Contact, No Sensitive Info Shared, Small Transfer, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure
Test 5	Account/Phone Takeover Risk, Suspicious Password Reset Protocol, Suspicious Link, Lack of Verification, Language/Voice Anomalies, Personal Info Disclosure, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure	Legitimate Password Reset Protocol, Caller-Initiated Contact, Legitimate Phone Number Assumed, No Personal Data/Money Request, Had similar experience before, Wrong Understanding/Unmatched Answer/Not Sure

Table 8

Analysis of each test accuracy by victimization status. Results are categorized by test category (fraudulent, legitimate, and ambiguous) and test type (phone-based and email-based), showing the counts of correct and incorrect responses for both victimized and non-victimized groups across 12 test scenarios. The asterisk (*) indicates statistical significance at $p < 0.05$.

Category	Test type	Test No.	Victimized		Non-victimized		Filtered	Overall		Chi-Square
			Incorrect	Correct	Incorrect	Correct		Incorrect	Correct	
Fraudulent	Phone-Based	2	49 (67%)	24 (33%)	10 (38%)	16 (62%)	18 (15%)	59 (60%)	40 (40%)	* ($\chi^2(1) = 6.5407, p = 0.0105$)
		4	24 (36%)	43 (64%)	7 (24%)	22 (76%)	21 (18%)	31 (32%)	65 (68%)	($\chi^2(1) = 1.2636, p = 0.2610$)
		6	24 (30%)	57 (70%)	2 (7%)	27 (93%)	7 (6%)	26 (24%)	84 (76%)	* ($\chi^2(1) = 6.1142, p = 0.0134$)
	Email-Based	7	2 (2%)	84 (98%)	0 (0%)	29 (100%)	2 (2%)	2 (2%)	113 (98%)	($\chi^2(1) = 0.6863, p = 0.4074$)
		10	17 (22%)	61 (78%)	4 (15%)	23 (85%)	12 (10%)	21 (20%)	84 (80%)	($\chi^2(1) = 0.6108, p = 0.4345$)
		12	12 (15%)	67 (85%)	3 (11%)	25 (89%)	10 (9%)	15 (14%)	92 (86%)	($\chi^2(1) = 0.3436, p = 0.5578$)
Legitimate	Phone-Based	3	30 (35%)	55 (65%)	9 (30%)	21 (70%)	2 (2%)	39 (34%)	76 (66%)	($\chi^2(1) = 0.2773, p = 0.5985$)
		8	10 (12%)	76 (88%)	3 (10%)	26 (90%)	2 (2%)	13 (11%)	102 (89%)	($\chi^2(1) = 0.0356, p = 0.8503$)
	Email-Based	9	26 (31%)	57 (69%)	12 (41%)	17 (59%)	5 (4%)	38 (34%)	74 (66%)	($\chi^2(1) = 0.9691, p = 0.3249$)
		11	26 (31%)	58 (69%)	8 (28%)	21 (72%)	4 (3%)	34 (30%)	79 (70%)	($\chi^2(1) = 0.1161, p = 0.7333$)
Category	Test type	Test No.	Victimized		Non-victimized		Filtered	Overall		Chi-Square
			Scam	Non-Scam	Scam	Non-Scam		Incorrect	Correct	
Ambiguous	Phone-Based	1	59 (71%)	24 (29%)	22 (71%)	9 (29%)	3 (3%)	81 (71%)	33 (29%)	($\chi^2(1) = 0.0001, p = 0.9903$)
		5	32 (41%)	47 (59%)	14 (48%)	15 (52%)	9 (8%)	46 (43%)	62 (57%)	($\chi^2(1) = 0.5237, p = 0.4693$)

Table 9
Association between each test accuracy and victimization status, assessed using logistic regression adjusting for four demographic control variables (age, gender, income, and education). Results are categorized by test category (fraudulent, legitimate, and ambiguous) and test type (phone-based and email-based), showing the counts of correct and incorrect responses for both victimized and non-victimized groups across 12 test scenarios. The asterisk (*) indicates statistical significance at $p < 0.05$.

Category	Test type	Test No.	Victimized		Non-victimized		Filtered	Overall		Logistic regression
			Incorrect	Correct	Incorrect	Correct		Incorrect	Correct	
Fraudulent	Phone-Based	2	48 (69%)	22 (31%)	8 (36%)	14 (64%)	25 (21%)	56 (61%)	36 (39%)	OR = 0.25 [0.08, 0.71], $p = 0.0098^*$
		4	22 (34%)	42 (66%)	5 (19%)	21 (81%)	27 (23%)	27 (30%)	63 (70%)	OR = 0.50 [0.16, 1.56], $p = 0.2301$
		6	23 (29%)	55 (71%)	2 (8%)	23 (92%)	14 (12%)	25 (24%)	78 (76%)	OR = 0.21 [0.04, 1.00], $p = 0.0496^*$
	Email-Based	7	2 (2%)	81 (98%)	0 (0%)	26 (100%)	8 (7%)	2 (2%)	107 (98%)	(zero cell in Vict $\times$ Test, <i>correctness</i>)
		10	16 (21%)	59 (79%)	3 (12%)	21 (88%)	18 (15%)	19 (19%)	80 (81%)	OR = 0.51 [0.13, 2.01], $p = 0.3360$
		12	12 (15%)	66 (85%)	3 (12%)	22 (88%)	14 (12%)	15 (15%)	88 (85%)	OR = 0.50 [0.12, 2.09], $p = 0.3391$
Legitimate	Phone-Based	3	29 (35%)	53 (65%)	8 (31%)	18 (69%)	9 (8%)	37 (34%)	71 (66%)	OR = 0.84 [0.31, 2.24], $p = 0.7223$
		8	10 (12%)	73 (88%)	3 (12%)	23 (88%)	8 (7%)	13 (12%)	96 (88%)	OR = 1.24 [0.27, 5.71], $p = 0.7867$
		9	25 (31%)	55 (69%)	11 (42%)	15 (58%)	11 (9%)	36 (34%)	70 (66%)	OR = 1.52 [0.55, 4.17], $p = 0.4192$
	Email-Based	11	26 (32%)	55 (68%)	6 (23%)	20 (77%)	10 (9%)	32 (30%)	75 (70%)	OR = 0.61 [0.20, 1.82], $p = 0.3748$
Category	Test type	Test No.	Victimized		Non-victimized		Filtered	Overall		Logistic regression
			Scam	Non-Scam	Scam	Non-Scam		Incorrect	Correct	
Ambiguous	Phone-Based	1	57 (71%)	23 (29%)	20 (74%)	7 (26%)	10 (9%)	77 (72%)	30 (28%)	OR = 0.96 [0.33, 2.77], $p = 0.9390$
		5	31 (40%)	46 (60%)	11 (44%)	14 (56%)	15 (13%)	42 (41%)	60 (59%)	OR = 1.21 [0.47, 3.15], $p = 0.6951$

Table 10
Mann-Whitney U Test results comparing victimized and non-victimized groups across 12 test scenarios, scenarios combined by test type (Phone-Based Legitimate/Fraudulent, Email-Based Legitimate/Fraudulent, and Ambiguous) and response tendencies (False Positive, False Negative). The asterisk (*) indicates statistical significance at $p < 0.05$.

Category	Test type	Test No.	Victimized	Non-victimized	Mann-Whitney U
Fraudulent	Phone-Based	Test 2	(n = 73, M = 2.19, SD = 1.12)	(n = 26, M = 2.69, SD = 1.14)	* ($U = 697.5, p = 0.0383$)
		Test 4	(n = 67, M = 3.12, SD = 1.41)	(n = 29, M = 3.38, SD = 1.42)	($U = 868.5, p = 0.4029$)
		Test 6	(n = 81, M = 3.21, SD = 1.43)	(n = 29, M = 3.97, SD = 1.10)	* ($U = 829.5, p = 0.0161$)
	Email-Based	Test 7	(n = 86, M = 4.48, SD = 0.80)	(n = 29, M = 4.76, SD = 0.50)	($U = 1045, p = 0.1123$)
		Test 10	(n = 78, M = 3.82, SD = 1.43)	(n = 27, M = 4.11, SD = 1.10)	($U = 971, p = 0.5199$)
		Test 12	(n = 79, M = 3.41, SD = 1.12)	(n = 28, M = 3.89, SD = 0.98)	($U = 842, p = 0.0512$)
Legitimate	Phone-Based	Test 3	(n = 85, M = 4.01, SD = 1.11)	(n = 30, M = 4.17, SD = 1.10)	($U = 1166.5, p = 0.4560$)
		Test 8	(n = 86, M = 4.41, SD = 0.85)	(n = 29, M = 4.45, SD = 0.67)	($U = 1267, p = 0.8869$)
	Email-Based	Test 9	(n = 83, M = 3.89, SD = 1.01)	(n = 29, M = 3.59, SD = 1.00)	($U = 1412, p = 0.1480$)
		Test 11	(n = 84, M = 3.87, SD = 1.12)	(n = 29, M = 3.90, SD = 1.09)	($U = 1206, p = 0.9368$)
Ambiguous	Phone-Based	Test 1	(n = 83, M = 2.11, SD = 0.99)	(n = 31, M = 2.13, SD = 0.83)	($U = 1230.5, p = 0.7079$)
		Test 5	(n = 79, M = 2.68, SD = 1.12)	(n = 29, M = 2.62, SD = 1.06)	($U = 1203, p = 0.6817$)
Fraudulent	Phone-Based	Test 2, 4, 6	(n = 86, M = 2.85, SD = 1.01)	(n = 31, M = 3.36, SD = 0.86)	* ($U = 909, p = 0.0086$)
Fraudulent	Email-Based	Test 7, 10, 12	(n = 86, M = 3.92, SD = 0.84)	(n = 29, M = 4.26, SD = 0.57)	($U = 965.5, p = 0.0678$)
Legitimate	Phone-Based	Test 3	(n = 85, M = 4.01, SD = 1.11)	(n = 30, M = 4.17, SD = 1.10)	($U = 1166.5, p = 0.4560$)
Legitimate	Email-Based	Test 8, 9, 11	(n = 86, M = 4.05, SD = 0.70)	(n = 29, M = 3.98, SD = 0.71)	($U = 1321.5, p = 0.6294$)
Ambiguous	Phone-Based	Test 1, 5	(n = 86, M = 2.35, SD = 0.83)	(n = 31, M = 2.37, SD = 0.71)	($U = 1332.5, p = 1.0000$)
False Negative	All types	All test cases	(n = 86, M = 1.49, SD = 1.26)	(n = 31, M = 0.84, SD = 1.02)	* ($U = 1746, p = 0.0081$)
False Positive	All types	All test cases	(n = 86, M = 1.07, SD = 1.01)	(n = 31, M = 1.03, SD = 1.09)	($U = 1393.5, p = 0.6968$)

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cose.2026.104961>.

Data availability

Data will be made available on request.

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